Model Assessment

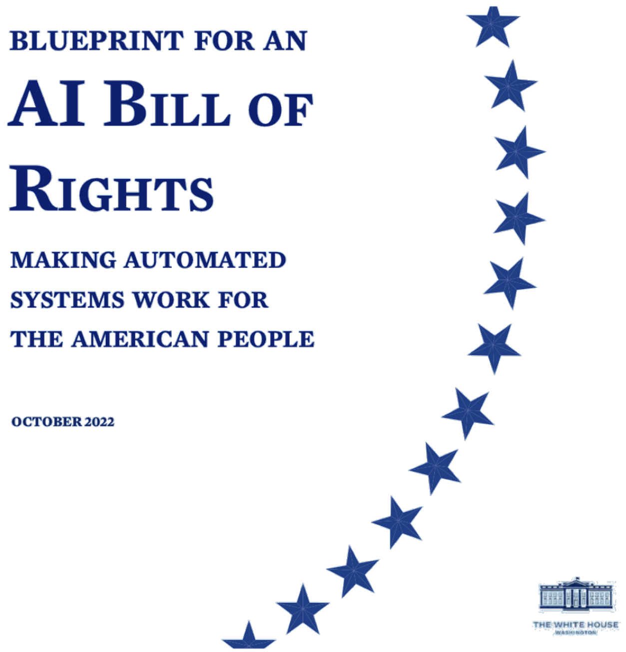
Dr. Ratna Babu Chinnam

Industrial & Systems Engineering

Wayne State University

US: AI Bill of Rights (October 2022)

- **SAFE AND EFFECTIVE SYSTEMS: You should be protected from unsafe or ineffective systems.** Automated systems should be developed with consultation from diverse communities, stakeholders, and domain experts to identify concerns, risks, and potential impacts of the system. Systems should undergo pre-deployment testing, risk identification and mitigation, and ongoing monitoring that demonstrate they are safe and effective based on their intended use, mitigation of unsafe outcomes including those beyond the intended use, and adherence to domain-specific standards. Outcomes of these protective measures should include the possibility of not deploying the system or removing a system from use. Automated systems should not be designed with an intent or reasonably foreseeable possibility of endangering your safety or the safety of your community. They should be designed to proactively protect you from harms stemming from unintended, yet foreseeable, uses or impacts of automated systems. You should be protected from inappropriate or irrelevant data use in the design, development, and deployment of automated systems, and from the compounded harm of its reuse. Independent evaluation and reporting that confirms that the system is safe and effective, including reporting of steps taken to mitigate potential harms, should be performed and the results made public whenever possible.
- **ALGORITHMIC DISCRIMINATION PROTECTIONS: You should not face discrimination by algorithms and systems should be used and designed in an equitable way.** Algorithmic discrimination occurs when automated systems contribute to unjustified different treatment or impacts disfavoring people based on their race, color, ethnicity, sex (including pregnancy, childbirth, and related medical conditions, gender identity, intersex status, and sexual orientation), religion, age, national origin, disability, veteran status, genetic information, or any other classification protected by law. Depending on the specific circumstances, such algorithmic discrimination may violate legal protections. Designers, developers, and deployers of automated systems should take proactive and continuous measures to protect individuals and communities from algorithmic discrimination and to use and design systems in an equitable way. This protection should include proactive equity assessments as part of the system design, use of representative data and protection against proxies for demographic features, ensuring accessibility for people with disabilities in design and development, pre-deployment and ongoing disparity testing and mitigation, and clear organizational oversight. Independent evaluation and plain language reporting in the form of an algorithmic impact assessment, including disparity testing results and mitigation information, should be performed and made public whenever possible to confirm these protections.
- **DATA PRIVACY: You should be protected from abusive data practices via built-in protections and you should have agency over how data about you is used.** You should be protected from violations of privacy through design choices that ensure such protections are included by default, including ensuring that data collection conforms to reasonable expectations and that only data strictly necessary for the specific context is collected. Designers, developers, and deployers of automated systems should seek your permission and respect your decisions regarding collection, use, access, transfer, and deletion of your data in appropriate ways and to the greatest extent possible; where not possible, alternative privacy by design safeguards should be used. Systems should not employ user experience and design decisions that obfuscate user choice or burden users with defaults that are privacy invasive. Consent should only be used to justify collection of data in cases where it can be appropriately and meaningfully given. Any consent requests should be brief, be understandable in plain language, and give you agency over data collection and the specific context of use; current hard-to- understand notice-and-choice practices for broad uses of data should be changed. Enhanced protections and restrictions for data and inferences related to sensitive domains, including health, work, education, criminal justice, and finance, and for data pertaining to youth should put you first. In sensitive domains, your data and related inferences should only be used for necessary functions, and you should be protected by ethical review and use prohibitions. You and your communities should be free from unchecked surveillance; surveillance technologies should be subject to heightened oversight that includes at least pre-deployment assessment of their potential harms and scope limits to protect privacy and civil liberties. Continuous surveillance and monitoring should not be used in education, work, housing, or in other contexts where the use of such surveillance technologies is likely to limit rights, opportunities, or access. Whenever possible, you should have access to reporting that confirms your data decisions have been respected and provides an assessment of the potential impact of surveillance technologies on your rights, opportunities, or access.
- **NOTICE AND EXPLANATION: You should know that an automated system is being used and understand how and why it contributes to outcomes that impact you.** Designers, developers, and deployers of automated systems should provide generally accessible plain language documentation including clear descriptions of the overall system functioning and the role automation plays, notice that such systems are in use, the individual or organization responsible for the system, and explanations of outcomes that are clear, timely, and accessible. Such notice should be kept up-to-date and people impacted by the system should be notified of significant use case or key functionality changes. You should know how and why an outcome impacting you was determined by an automated system, including when the automated system is not the sole input determining the outcome. Automated systems should provide explanations that are technically valid, meaningful and useful to you and to any operators or others who need to understand the system, and calibrated to the level of risk based on the context. Reporting that includes summary information about these automated systems in plain language and assessments of the clarity and quality of the notice and explanations should be made public whenever possible.
- **HUMAN ALTERNATIVES, CONSIDERATION, AND FALLBACK: You should be able to opt out, where appropriate, and have access to a person who can quickly consider and remedy problems you encounter.** You should be able to opt out from automated systems in favor of a human alternative, where appropriate. Appropriateness should be determined based on reasonable expectations in a given context and with a focus on ensuring broad accessibility and protecting the public from especially harmful impacts. In some cases, a human or other alternative may be required by law. You should have access to timely human consideration and remedy by a fallback and escalation process if an automated system fails, it produces an error, or you would like to appeal or contest its impacts on you. Human consideration and fallback should be accessible, equitable, effective, maintained, accompanied by appropriate operator training, and should not impose an unreasonable burden on the public. Automated systems with an intended use within sensitive domains, including, but not limited to, criminal justice, employment, education, and health, should additionally be tailored to the purpose, provide meaningful access for oversight, include training for any people interacting with the system, and incorporate human consideration for adverse or high-risk decisions. Reporting that includes a description of these human governance processes and assessment of their timeliness, accessibility, outcomes, and effectiveness should be made public whenever possible.



Source: LINK

Motivation

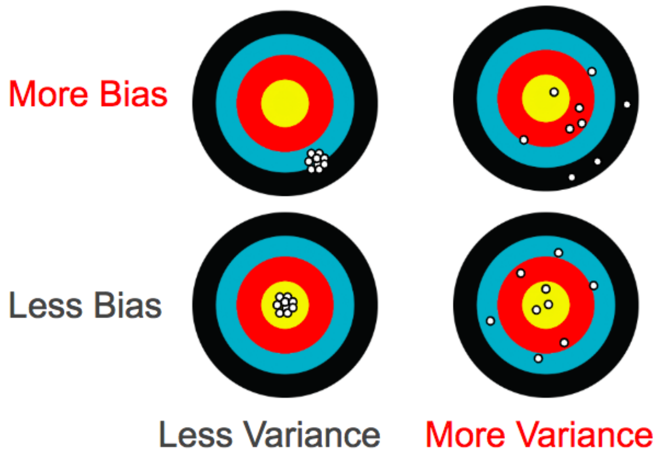
Google Says It Will Address AI, ML Model Bias with Technology called TCAV

By Larry Dignan | May 7, 2019 | ZDNet | Link

- Google CEO Sundar Pichai said the company is working to make its AI and ML models more transparent to defend against bias.
- Pichai outlined several AI enhancements, but the bigger takeaway for developers and data scientists may be something called TCAV.
- Testing with Concept Activation Vectors (TCAV) is an interpretability method to help us understand signals driving neural network predictions.
- In theory, TCAV's ability to understand signals could surface bias because it would highlight whether males were a signal over females and surface other issues such as race, income and location. Using TCAV, one can see how high value concepts are valued.

Judging Model Performance & Robustness

- Accuracy isn't Everything
  - Timeliness / Lead time to action
  - Class imbalances / Bias
  - Cost differences: False +ve vs False –ve
    - Requires cost-sensitive learning
- Performance Repeatability
  - Ideal conditions vs real-world
- Variable Selection/Influence Consistency
- False Alarms Fatigue!
- Purely Data-Driven vs Mechanistic/Physics Based Models
- Ensembles: Combine Approaches for Robustness



Real-life Examples of ML Discrimination

WAYNE STATE UNIVERSITY

- **Racial Bias in US Health Care Risk Algorithm** (Vartan 2019 | Link)
  - Algorithm used on > 200M people in US hospitals to predict which patients would need extra medical care heavily favored white patients
    - Race itself wasn't a variable used in this algorithm, however, another variable highly correlated to race was, healthcare cost history.
    - For various reasons, black patients incurred lower health-care costs than white patients with the same conditions on average.
  - Optum helped reduce the level of bias by 80%.
- **AI Bias in US COMPAS Algorithm** (Larson et al., 2016 | Propublica)
  - Correctional Offender Management Profiling for Alternative Sanctions
  - Used in US courts to predict likelihood that a defendant becomes a recidivist
  - Model predicted twice as many false positives for black offenders (45%)
- **Amazon's Sexist Hiring Algorithm** (Dastin 2018 | Reuters)
  - Models trained to vet applicants by patterns in résumés from 10-years
  - System taught itself that male candidates were preferable.
    - It penalized résumés that included the word "women's", as in "women's chess club captain". And it downgraded graduates of two all-women's colleges.
  - Amazon edited the programs to make them neutral to these terms. But that was no guarantee that the machines would not devise other ways of sorting candidates that could prove discriminatory, the people said.
  - Company ultimately disbanded the team because executives lost hope

Source: Shin (2020) Real-life Examples of Discriminating AI| Link

Model Assessment

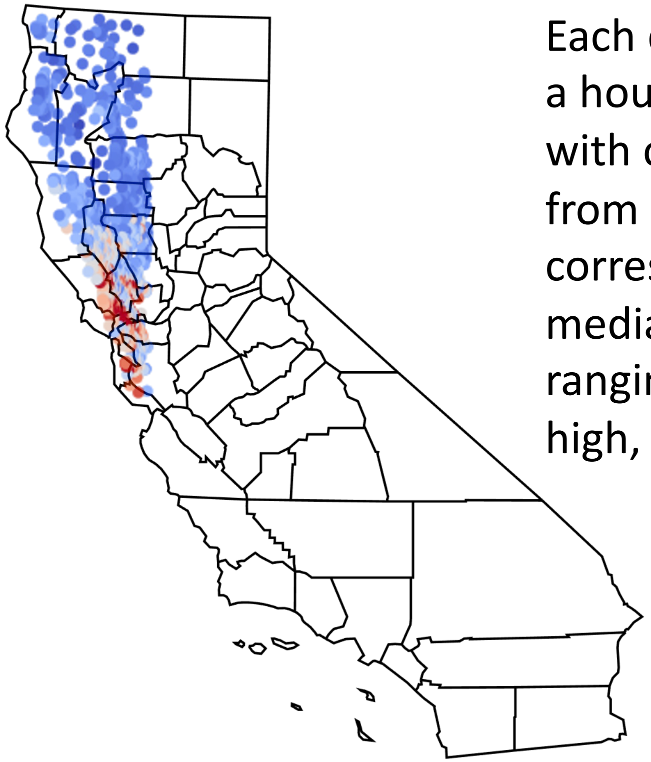
- First Requirement: Dataset offers (best) features to carry out task
- Feature selection/extraction might be necessary
- Small datasets: *K*-fold cross-validation strategy and/or regularization
- Build a model that best addresses the *bias-variance dilemma*
  - For “small” datasets, as one decreases bias, variance increases (and vice versa)
- Effective Model:
  - Selecting most appropriate modeling approach
  - Optimize model complexity (e.g., in MLPs, # layers, # nodes/layer), model parameters (e.g., transfer function)
  - Appropriate learning algorithm and parameters
- Vigorous Testing & Sensitivity Analysis

Some Sources of Model Bias

- Missing Feature Values
  - If your data set has features that have missing values for many examples, that could be an indicator that certain key characteristics of your data set are under-represented.
- Unexpected Feature Values
- Data Skew
  - Geographical Bias Example: If this unrepresentative sample were used to train a model to predict California housing prices statewide, the lack of housing data from southern portions of California would be problematic.

	longitude	latitude	total_rooms	population	households	median_income	median_house_value
count	17000.0	17000.0	17000.0	3000.0	3000.0	3000.0	17000.0

California Housing Dataset



Each dot represents a housing block, with colors ranging from blue to red corresponding to median house price ranging from low to high, respectively.

Evaluating for Bias: Example

- Consider a model developed to predict presence of tumors
  - 500 records each from female and male patients

- Calculated metrics separately for female and male patients show stark differences

Confusion Matrix

Ground Truth

Prediction

True Positives (TPs): 16	False Positives (FPs): 4
False Negatives (FNs): 6	True Negatives (TNs): 974

"Precision"

"Recall"

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{16}{16 + 4} = 0.800$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{16}{16 + 6} = 0.727$$

Female Patient Results

True Positives (TPs): 10	False Positives (FPs): 1
False Negatives (FNs): 1	True Negatives (TNs): 488

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{10}{10 + 1} = 0.909$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{10}{10 + 1} = 0.909$$

Male Patient Results

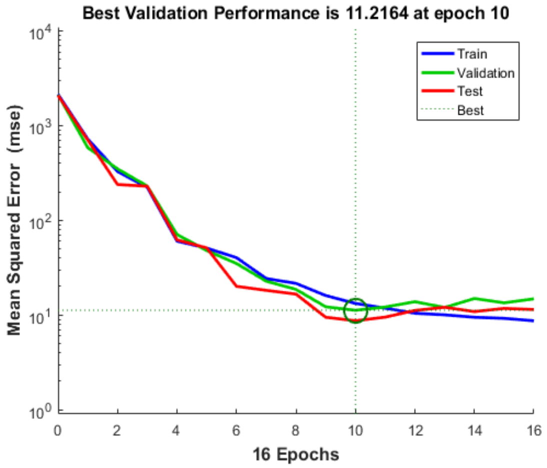
True Positives (TPs): 6	False Positives (FPs): 3
False Negatives (FNs): 5	True Negatives (TNs): 486

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{6}{6 + 3} = 0.667$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{6}{6 + 5} = 0.545$$

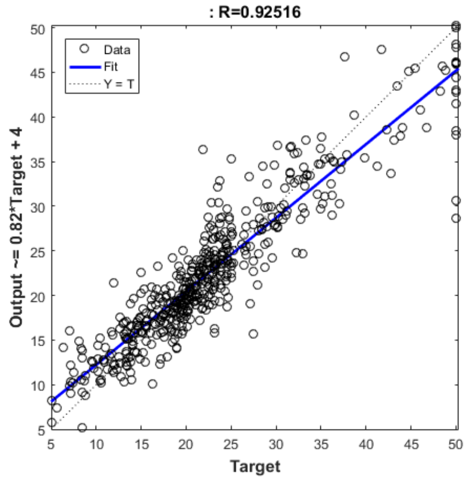
Function Approximation & Regression

- Performance criterion for learning model should be a "complete" measure
  - Accuracy and precision (e.g., MSE and MAD are complete measures but not $E(error)$)
- Model errors across training, validation, and testing datasets should be consistent
- Regression plots are useful
  - For training, validation, and testing datasets
  - Ideally have a high R^2 and an intercept of zero and slope of 1



Sample Cross-Validation MSE Plot

`Matlab:-> [net,tr] = train(net,x,t); plotperform(tr)`

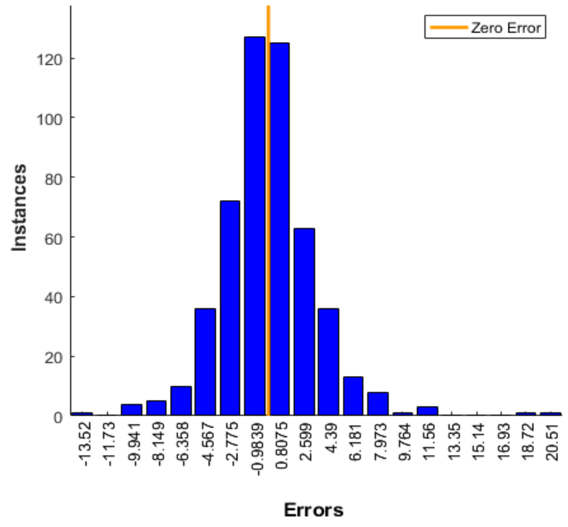


Sample Plot of Actual Vs. Predicted

`Matlab:-> y = net(x); plotregression(t,y)`

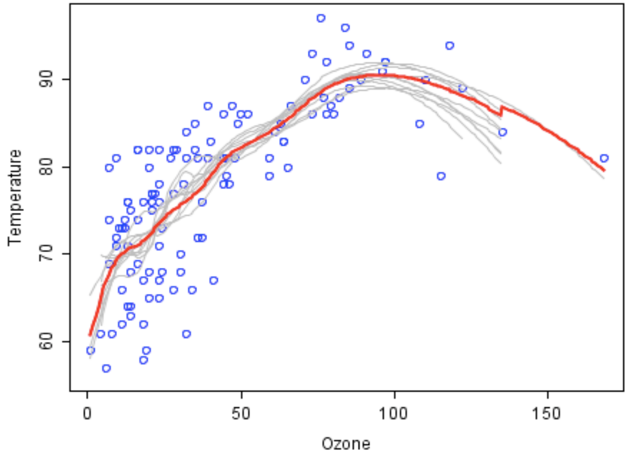
Function Approximation & Regression …

- Error histograms are also quite useful
  - Most errors should be close to zero
  - Significant outliers (large errors) should be investigated further
    - Data collection errors, missing variables
- Track performance across replications
  - Many machine learning models can get stuck in local optimum solutions
  - Training process must be repeated several times to demonstrate that results are consistent



Sample residual error histogram

`Matlab:-> e = t - y; ploterrhist(e)`

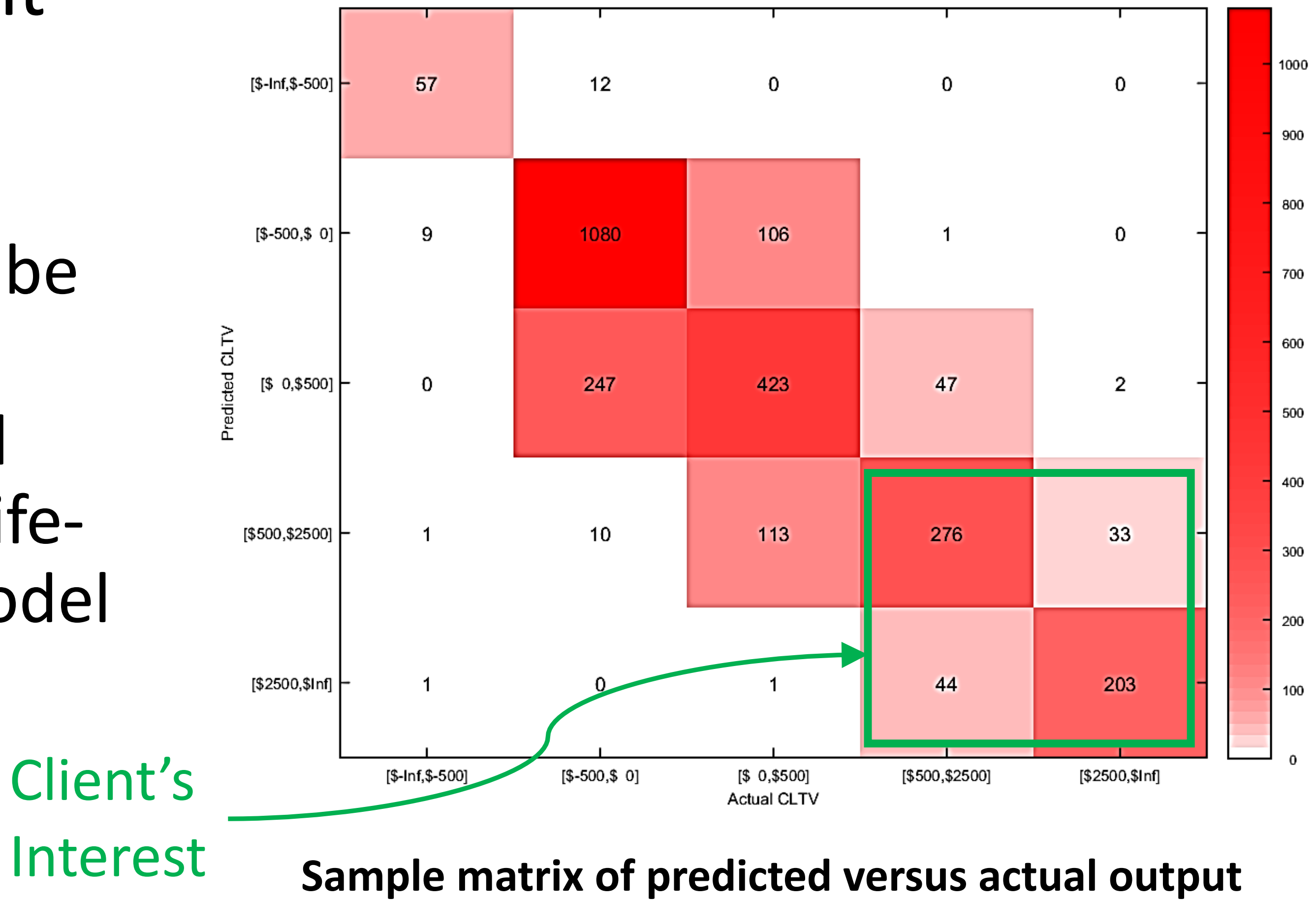


Results from different runs (gray lines)

`Matlab:-> y = net(x); plotregression(t,y)`

Function Approximation & Regression …

- Binning errors by range might be of interest for certain applications
  - Certain types of errors might be more important
  - Example: Table reports actual versus estimated Customer Life-Time Values (CLTV) from a model



Sample matrix of predicted versus actual output

Pattern Recognition (Classifiers)

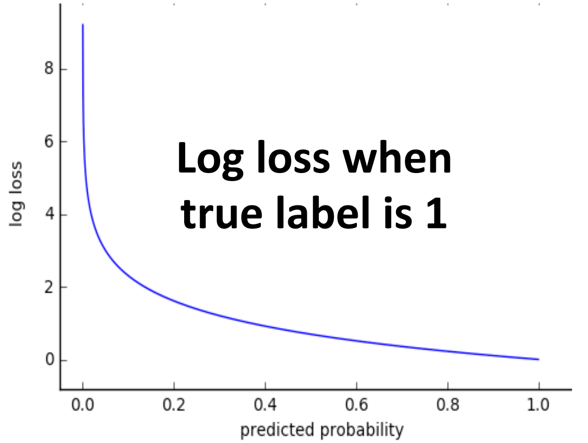
- **Cross-entropy loss** appropriate for training classifiers (Link1, Link2)
  - Penalizes extremely inaccurate outputs (p near $1 - t$) with little penalty for fairly correct classifications (p near t)

$$\text{CE Loss} = -\sum_{c=1}^{M} t_{o,c} \log_{10}(p_{o,c})$$

M: number of classes

$t_{o,c}$: Binary target (0 or 1) if label c is correct for observation o

$p_{o,c}$: predicted probability observation o is of class c



- Example: Let us take a three-class classification problem
  - True Output (Target) for a particular observation: [0 0 1]
  - Predicted Output for this observation: [0.04 0.13 0.83]

$$\text{CE Loss} = -\sum_{c=1}^{M} y_{o,c} \log(p_{o,c})$$

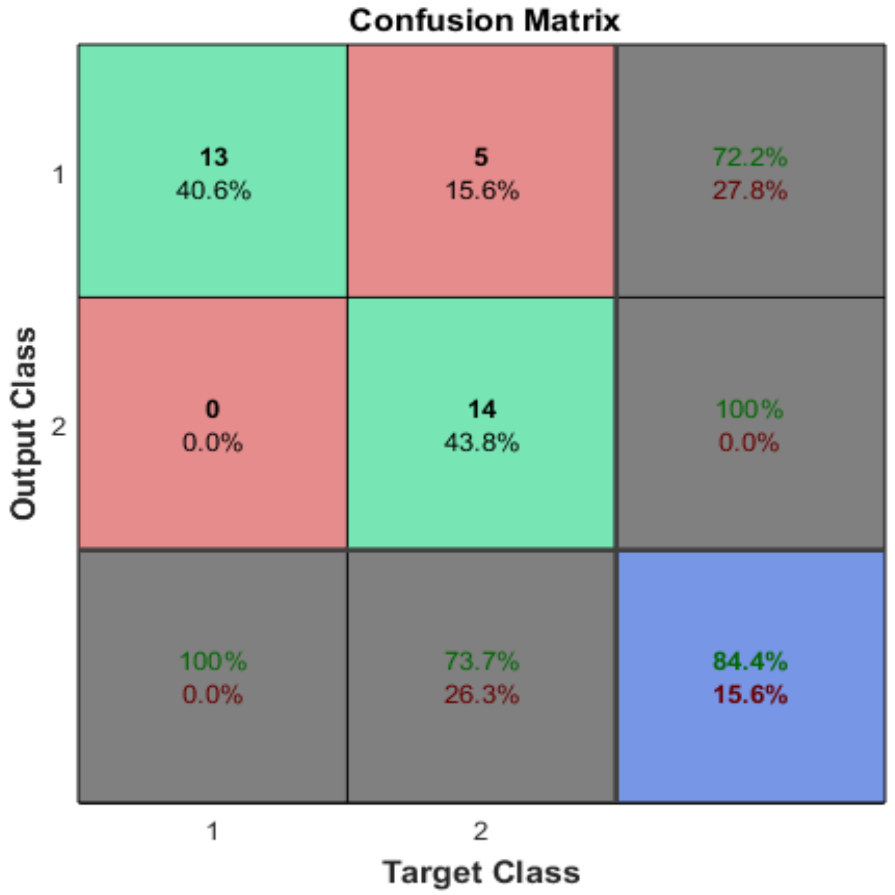
$$= -[0 \times \log(0.04) + 0 \times \log(0.13) + 1 \times \log(0.83)] = -\log(0.83) = 0.081$$

Distinguish multi-class classification from multi-label classification!

A sample can belong to more than 1 class.

Pattern Recognition (Classifiers): Confusion Matrix

- An effective performance measure is "***confusion matrix***"
  - Shows %s of correct and incorrect classifications
  - Diagonal entries are correct (green)



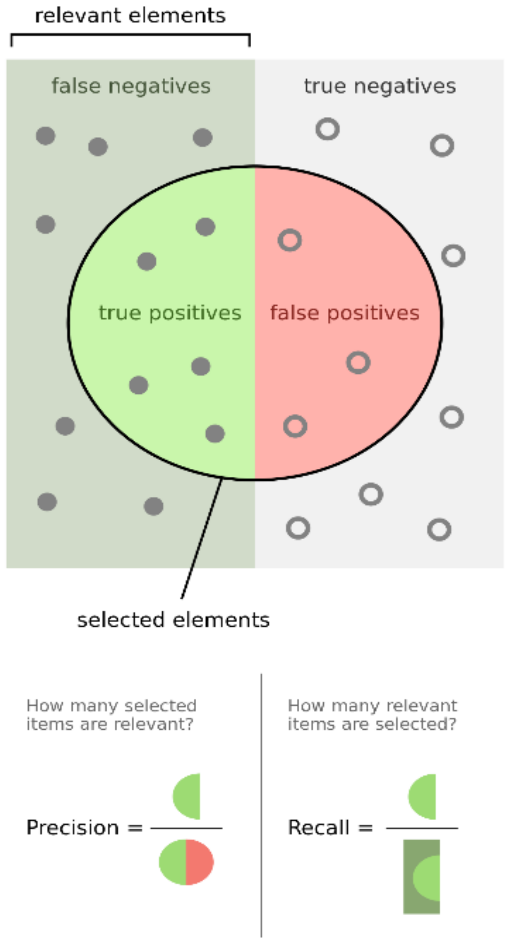
Sample Confusion Matrix for Testing Dataset With Two Target Classes

```
Matlab:-> [net,tr] = train(net,x,t);
testX = x(:,tr.testInd); testT = t(:,tr.testInd); <- Testing Observations
testY = net(testX); plotconfusion(testT,testY)
```

Pattern Recognition (Classifiers): Confusion Matrix

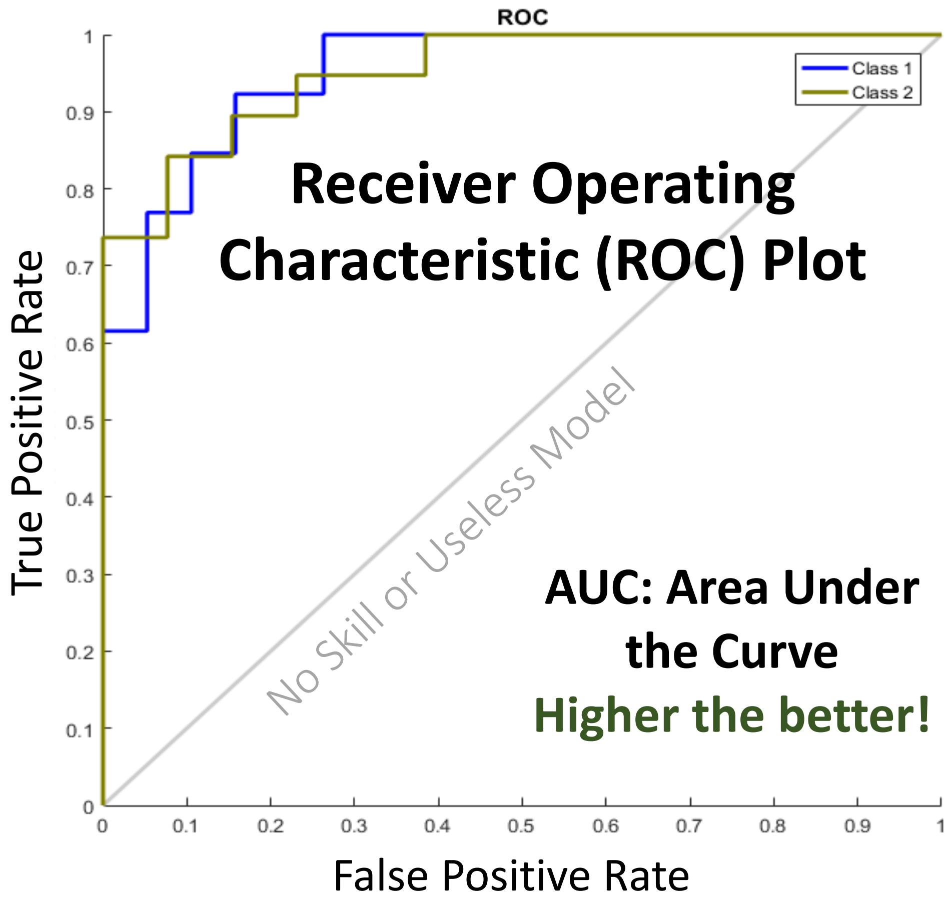
Binary Classification Example

		Predicted Condition		
		Predicted +ve	Predicted −ve	
True Condition	Condition +ve	True^{+}	False^{-} (Type-II error)	**Sensitivity** or **Recall**= $\frac{\sum True^{+}}{\sum Condition^{+}}$
True Condition	Condition −ve	False^{+} (Type-I error)	True^{-}	**Specificity** = $\frac{\sum True^{-}}{\sum Condition^{-}}$
	Prevalence $= \frac{\sum Condition^{+}}{\sum Total}$	**Precision** = $\frac{\sum True^{+}}{\sum Predicted^{+}}$	**-ve Predictive Value** $= \frac{\sum True^{-}}{\sum Predicted^{-}}$	**Accuracy** = $\frac{\sum True^{+} + \sum True^{-}}{\sum Total}$



Pattern Recognition (Classifiers): ROC Plot

- Shows how false +ve and true +ve rates change as “threshold” for output classification is varied from 0 to 1
  - Straightforward for binary classification
- Ideally, we are looking for the curve to bow towards the top left corner
  - A completely random model with no utility will yield an ROC plot that is diagonal at 45%
- One should pick a threshold that best balances the two rates to meet the needs of the application
- For multi-class problems, each class will be compared against all other classes
  - One ROC curve for each class (not effective)



Sample ROC Curve Plot for Binary Classification Problem (Cancer Detection)
(Class 1: cancer patients; class 2: normal patients)
`Matlab:-> plotroc(testT,testY)`

Pattern Recognition (Classifiers): Class Imbalance

- **Class Imbalance**: Observations not uniformly distributed across classes
  - Example: Outcome of "strep test" for patients with sore throat condition is mostly "negative" for Streptococcal pharyngitis
- $\boldsymbol{F_1}$ **Score**: Option for binary classification with class imbalance

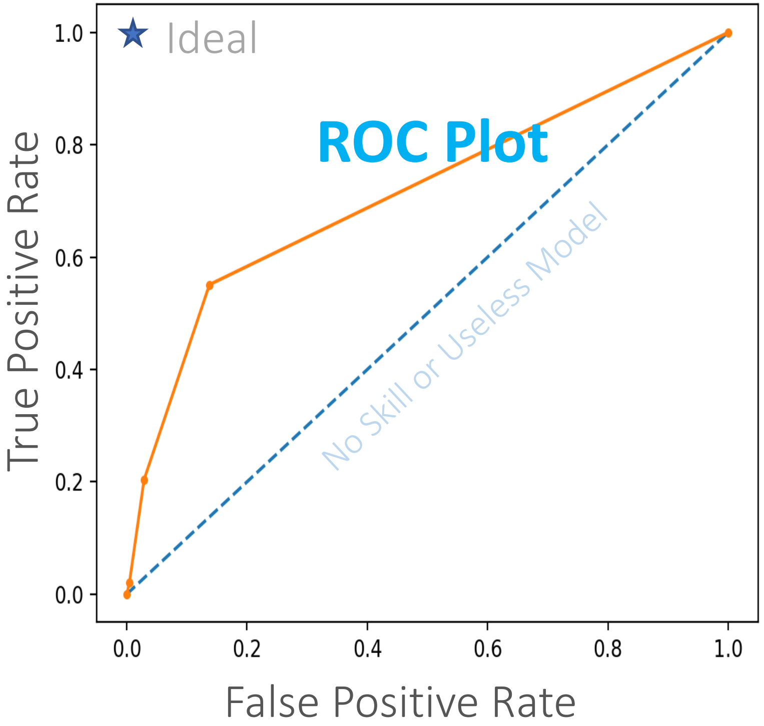
$$F_1\ Score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

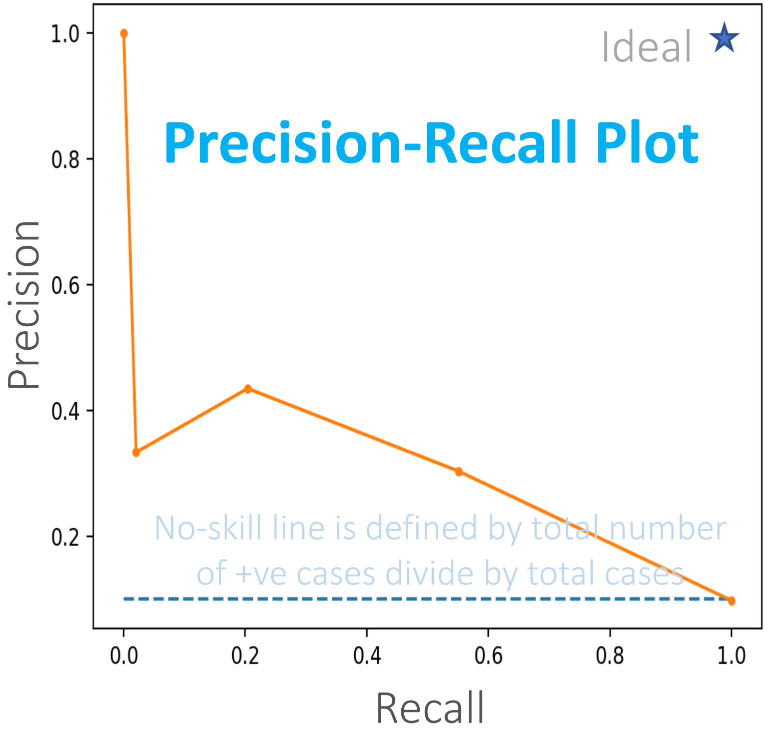
- $\boldsymbol{F_1}$ **Score vs. Accuracy**: Weather Prediction Example
  - Suppose only 10% of days are "sunny"
  - A model that (almost) always predicts "not sunny" (say 99.999% of the time, randomly) has an accuracy of 90% whereas it's F_1 score will be 0
    - Precision will be $\approx 0/(0+0)$ NaN and Sensitivity will be $0/(0+10) = 0$

Pattern Recognition (Classifiers): Precision-Recall Plot

- ROC plot not that effective under class imbalance
- Precision-Recall curve is more useful
  - Does not make use of true negatives; only concerned with correct prediction of minority class
- For more discussion, check out this link.

This model is useful as suggested by ROC curve. However, it is mostly useful for making few true +ve predictions and there are a lot of false –ve predictions.





ROC and Precision-Recall Curves for a Weak Model Learnt from a Highly Imbalanced Dataset (Source: Link)

Tactics for Addressing Class Imbalance

- Collect more data
- Try different algorithms
- Change performance metric during model optimization (e.g., F_1 Score)
- Resample your dataset (Link)
  - Over-sampling: Add copies of instances from under-represented class
  - Under-sampling: Delete instances from over-represented class
- Generate synthetic examples
  - Synthetic Minority Over-sampling Technique (SMOTE): Creates synthetic samples from minor class by selecting two or more similar instances (using a distance measure) and perturbing an instance one attribute at a time by a random amount within the difference to the neighboring instances (Python Module | R DMwR)
- Cost-sensitive learning: Add more cost for minority classes
- Try different perspective: Anomaly detection (e.g., 1-class classification)
- Try getting creative: Redefine the problem or classes!

Pattern Recognition (Classifiers): Cost Sensitivities

- **Setting**: In cases where the cost of false +ves is different from cost of false –ves, "***cost sensitive learning***" will be necessary
  - Example: Cost of a false negative for cancer screening could be a 1,000 times higher than a false positive
- **Vigorous Approach**: Objective for the learning model should explicitly account for differences in costs of different error types
- **Heuristic Approach**: (Binary Classification Example)
  - Model learnt giving equal weight for both types of errors
  - Pick a classifier (threshold) by selecting a point on the ROC that minimizes overall cost

REFERENCE: Elkan (2001) Foundations of Cost-Sensitive Learning | PDF

Time-Series Forecasting

- Partition time-series as consecutive segments for training, validation, and testing (with no overlaps) to demonstrate generalization ability
- MSE plots during training are meaningful as well
- Some popular performance measures:
  - ***MAPE (Mean Absolute Percent Error)***:
    - MAPE is scale sensitive and should not be used with low-scale data
      - MAPE will often take on extreme values for small changes
    - MAPE is undefined when Actual value is zero (enters denominator)
  - ***MAD (Mean Absolute Deviation)***:
    - Calculated as average of unsigned errors

$$\left(\frac{1}{n}\sum\frac{|Actual - Forecast|}{|Actual|}\right)*100$$

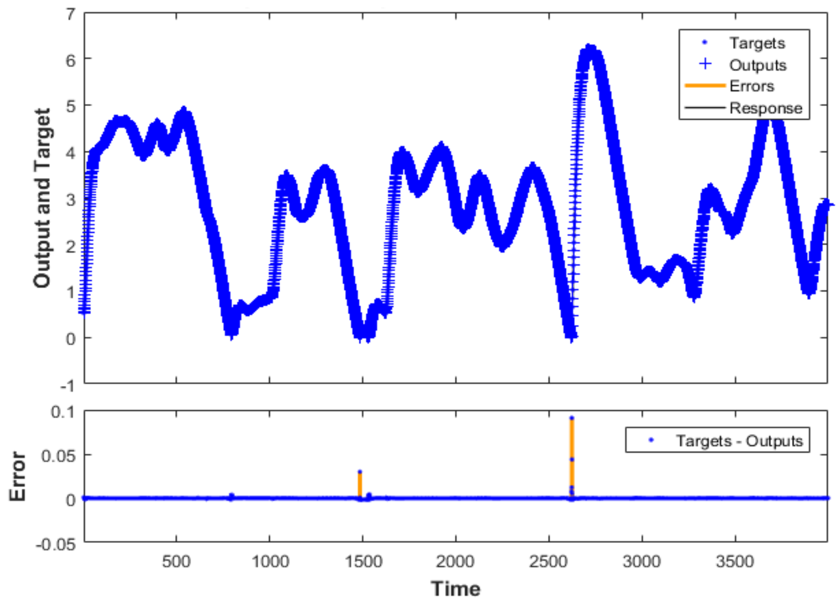
Month	Actual	Forecast	Absolute Percent Error
1	112.3	124.7	11.0%
2	108.4	103.7	4.3%
3	148.9	116.6	21.7%
4	117.4	78.5	33.1%
MAPE			**17.6%**

$$\frac{1}{n}\sum|Actual - Forecast|$$

Month	Actual	Forecast	Absolute Error
1	112.3	124.7	12.4
2	108.4	103.7	4.7
3	148.9	116.6	32.3
4	117.4	78.5	38.9
MAD			**22.08**

Time-Series Forecasting …

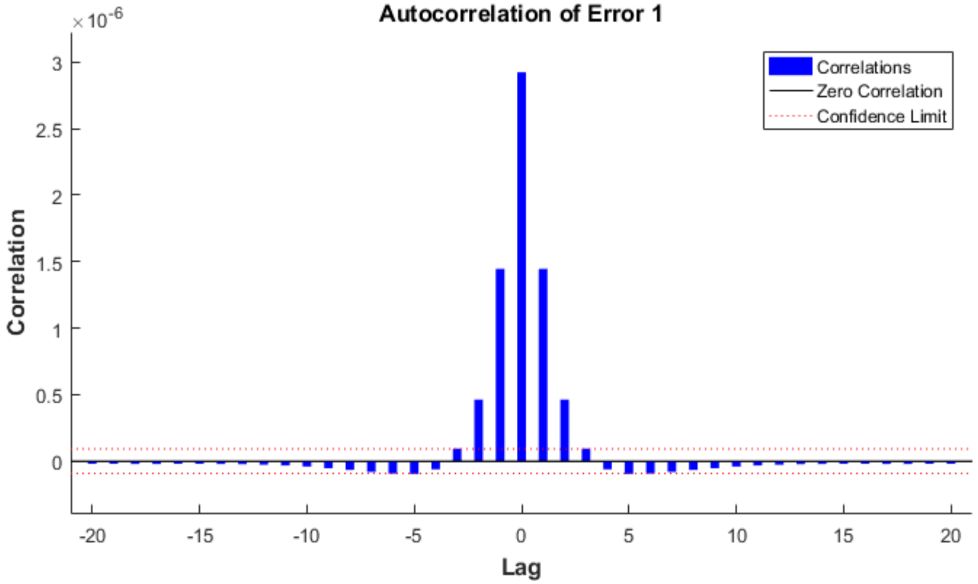
- Plot actual vs predicted values for testing times-series segment along with errors



Sample Forecasting Plot

`Matlab:-> plotresponse(Ts,Y)`

- Plot correlation of error at time t, $e(t)$ with errors over varying lags, $e(t + lag)$
  - All lines should be short



Autocorrelation of Error

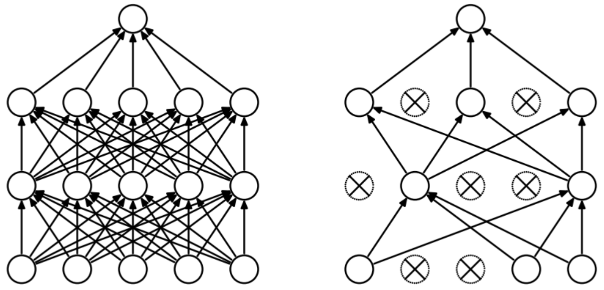
`Matlab:-> E = gsubtract(Ts,Y); ploterrcorr(E)`

- Model should outperform a naïve model that predicts $\hat{Y}(n+1) = Y(n)$
- Test accuracy of "recursive" predictions if relevant

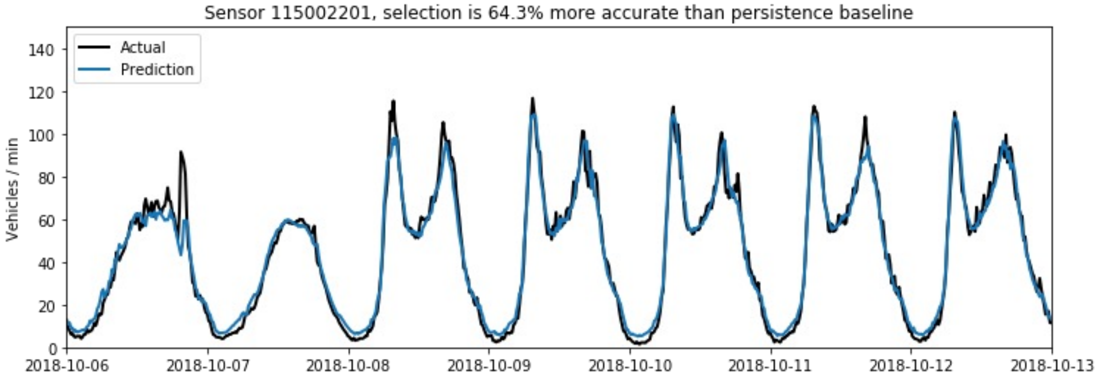
Generating Confidence Intervals using “Dropout” Feature

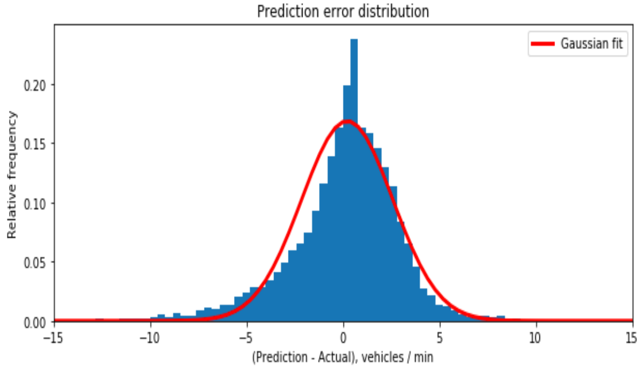
WAYNE STATE UNIVERSITY

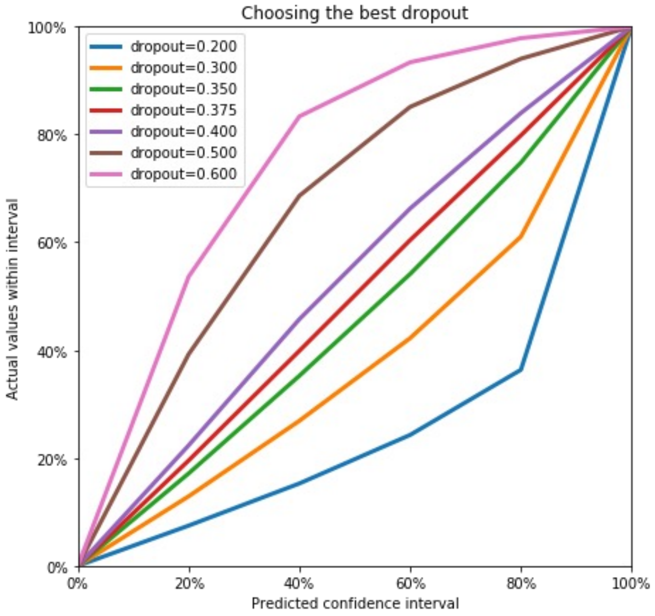
- To get deterministic ANN models to return confidence intervals, exploit “dropout” feature during inference
  - Dropout “turns off” a randomly chosen fraction of model neurons during training to prevent overfitting

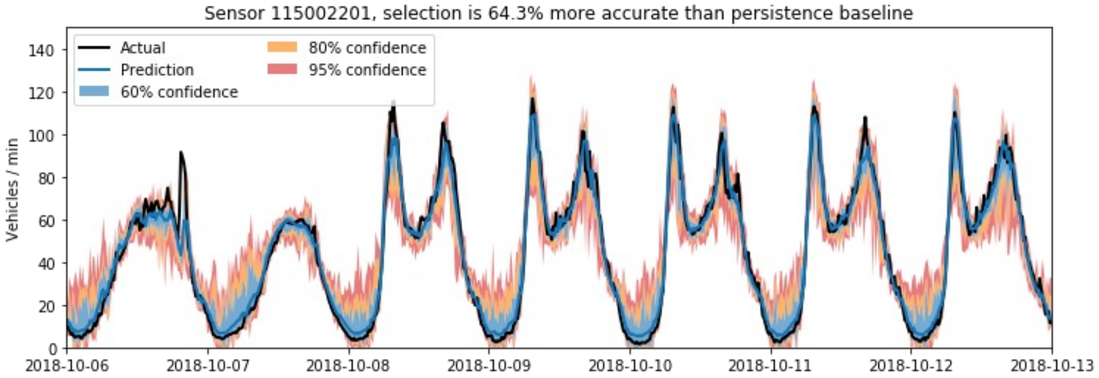


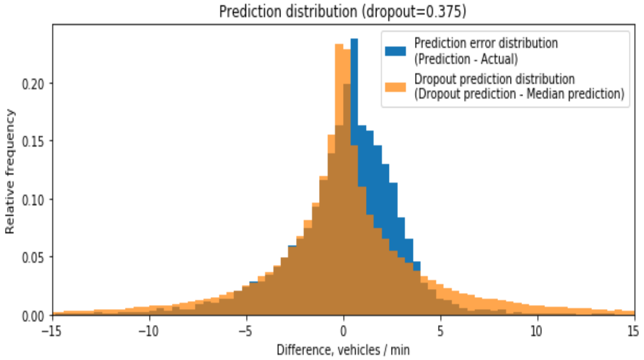
(a) Standard Neural Net (b) After applying dropout.







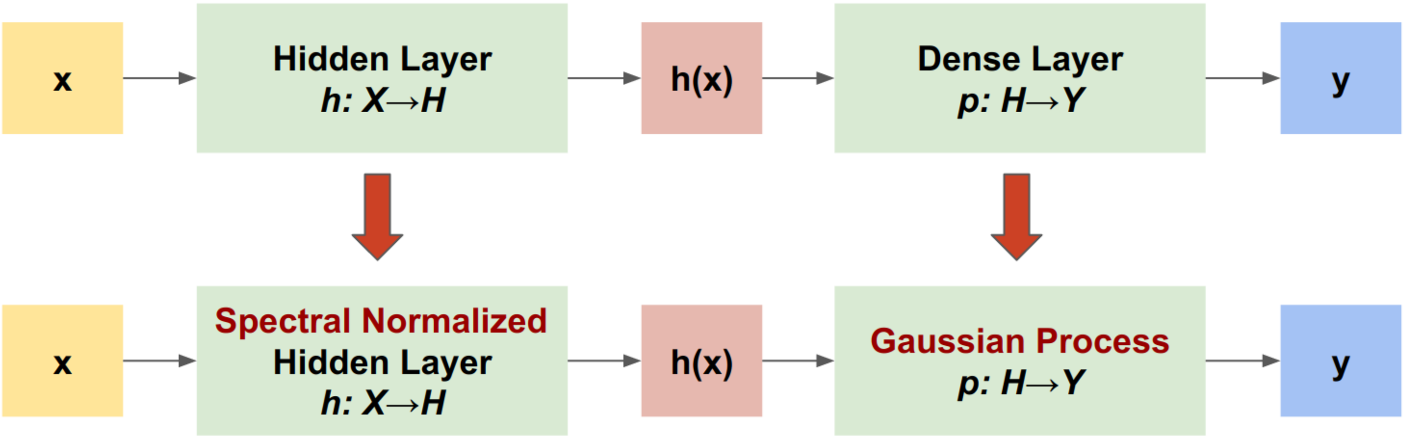




Source: Blake (2019) How to generate neural network confidence intervals with Keras | [Link]

Uncertainty-Aware Learning with Gaussian Process

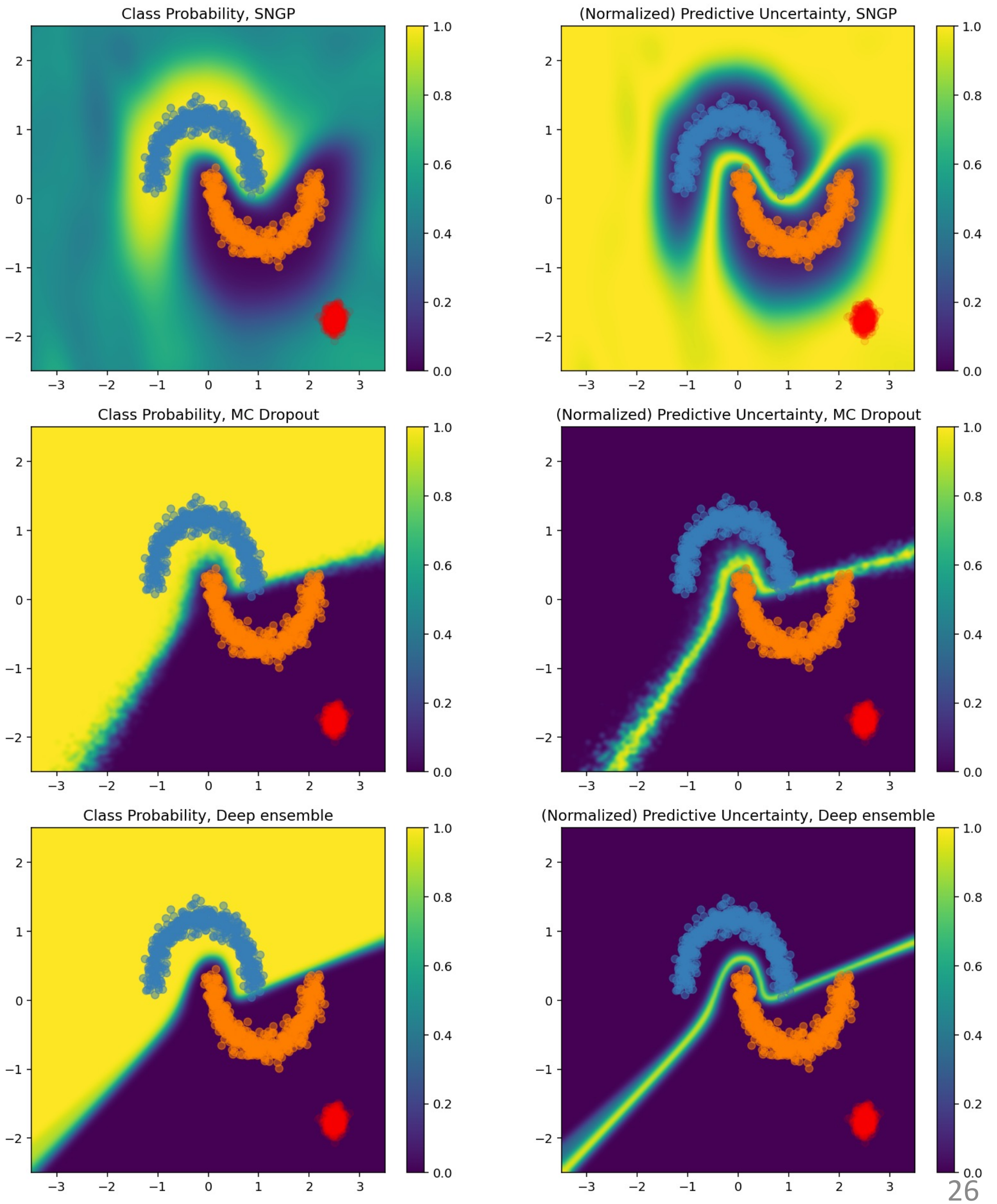
- Quantifies uncertainty using Spectral-normalized Neural Gaussian Process
- SNGP improves a deep learning classifier's ***distance awareness*** by applying simple modifications to the network.
  - It is a measure of how its predictive probability reflects the distance between test example and training data.
- Given a deep network, SNGP makes two simple changes to the model:
  - It applies spectral normalization to the hidden residual layers.
  - It replaces the Dense output layer with a Gaussian process layer.



Source: Google Tensorflow | Link

Uncertainty-Aware Learning with Gaussian Process

- SNGP has several advantages over other approaches (e.g., Monte Carlo dropout or Deep ensemble):
  - Works for a range of state-of-the-art residual-based architectures (e.g., ResNet, DenseNet, or BERT).
  - It is a single-model method—does not rely on ensemble averaging. Therefore, can be scaled to large datasets like ImageNet and Jigsaw Toxic Comments classification.
  - It has strong out-of-domain detection performance due to the distance-awareness property.
- Downsides of this method are:
  - Predictive uncertainty of SNGP is computed using Laplace approximation. Theoretically, posterior uncertainty of SNGP is different from that of a GP.
  - SNGP training needs a covariance reset step at the beginning of a new epoch. This can add a tiny amount of extra complexity to a training pipeline.
- Tensorflow tutorial shows how to implement this using Keras callbacks: LINK



Source: Google Tensorflow | Link